



PCTE INSTITUTE OF ENGINEERING & TECHNOLOGY

BTECH 2023 A

INTERNSHIP PROJECT SYNOPSIS

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1) Title of Project

PlayConnect: A Real-Time Sports Matchmaking and Community Networking Application

2) Objectives & Significance of Project

Objectives

The main objective of PlayConnect is to help people find nearby players and easily organize sports matches.

The main objectives are:

- **Connect Players:** To help users find nearby people who are interested in the same sports, such as cricket, football, basketball, badminton, etc.
- **Find Nearby Matches:** To show users sports matches happening near their location.
- **Create Matches:** To allow users to create a match by adding the sport, location, date, and time.
- **Join Matches:** To allow users to see available matches and join the matches they are interested in.

- Chat with Players: To provide a chat feature so players can communicate and plan their games without sharing their personal phone numbers.
- Build a Sports Community: To allow sports lovers to connect, share sports-related content, and interact with other players.
- Promote Physical Activity: To encourage people to play outdoor sports and maintain an active lifestyle.

Significance

Finding enough players for a sports game can be difficult, especially when a person does not have a regular sports group. People often use WhatsApp or other social media groups, but these groups can become difficult to manage.

PlayConnect solves this problem by providing a single platform where users can find nearby players, create matches, join games, and communicate with other sports lovers.

The application uses location services to help users find matches near them, making it easier to connect with local players.

3) Methodologies / Algorithms / Tools / Techniques

The PlayConnect application is developed using modern technologies for mobile application development and real-time communication.

React Native & Expo

React Native is used to create the mobile application.

Expo is used to make the development and testing of the React Native application easier.

Using React Native allows us to build the application using JavaScript and React.

Convex

Convex is used as the backend and database of PlayConnect.

It is used to store and manage information such as:

- User information
- Match details
- Joined matches
- Chat messages
- Community data

Convex also provides real-time updates, so changes such as new messages or match updates can appear in the application quickly.

Geolocation and Distance Calculation

Location is an important part of PlayConnect.

The application gets the user's location using the device's GPS and calculates the distance between the user and a match.

The basic process is:

User Location → Match Location → Calculate Distance → Check Radius → Show Nearby Matches

The geolib library is used to calculate the distance.

For example, if the selected radius is 5 km, the application can show matches that are within 5 km of the user's location.

React Context API

React Context API is used to manage information that is needed in different parts of the application.

For example, it can be used for:

- User session
- Theme settings
- Application preferences

4) Results

The PlayConnect application was successfully developed and tested on mobile devices.

The application provides features for finding nearby sports matches, creating and joining matches, chatting with players, and interacting with the sports community.

Main Results

Feature	Result
Nearby Matches	Users can find matches near their location.
Create Match	Users can create a match with sport, date, time, and location.
Join Match	Users can join available matches.
Chat	Players can communicate with each other.

Feature	Result
Location Filtering	Matches can be filtered according to distance.
Community	Users can interact with other sports lovers.
Real-Time Updates	Match and chat information can be updated quickly.

5) Conclusion

PlayConnect successfully provides a solution for people who have difficulty finding players and organizing sports matches.

The application combines nearby match discovery, match creation, joining matches, chat, and community features in one platform.

By making it easier to find nearby players and sports activities, PlayConnect can encourage people to participate more in outdoor sports and maintain an active lifestyle.

The project also shows how technologies such as React Native, Expo, Convex, GPS, and real-time data can be used to build a useful mobile application.

6) Limitations of Project & Future Scope

Limitations

- ❖ 1. Internet Requirement
- ❖ 4. Location Permission

Future Scope

- ❖ 1. Turf and Ground Booking
- ❖ 2. AI-Based Matchmaking
- ❖ 3. Tournaments and Leaderboards
- ❖ Player Statistics